

-ASSIGNMENT NO. 2

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TITLE: Inclusion and Exclusion Principle

finding student participation in Sports Club and Music Club in a university survey. The university has 200 students, and the data collected from the survey is as follows:

- 120 students are members of the Sports Club.
- 90 students are members of the Music Club.
- 50 students are members of both the Sports Club and the Music Club.

*/

```
#include<iostream>
```

```
using namespace std;
```

```
int main(){
```

```
    int students , sports, music , both;
```

```
    cout <<"enter total no. of students :"<<endl;
```

```
    cin >> students;
```

```
    cout <<"enter no. of students in sports club :"<<endl;
```

```
    cin >> sports;
```

```
    cout <<"enter no. of students in music club :"<<endl;
```

```
    cin >> music;
```

```
    cout <<"enter no. of students in both clubs:"<<endl;
```

```
    cin >> both;
```

```
    //students in only sports club (A-B)
```

```
    int onllys = sports-both;
```

```
    //students in music club
```

```
    int onlym = music-both;
```

```
    //students in atleast one club(union)
```

```

int atleast= onlys+onlym+both;
//students in neither club (U-union)
int neither = students - atleast;

cout << "students : " << students << endl;
cout << "students in sports club : " << sports << endl;
cout << "students in music club : " << music << endl;
cout << "students in both clubs : " << both << endl;
cout << "students in only sports club (A-B) : " << onlys << endl;
cout << "students in only music club (B-A) : " << onlym << endl;
cout << "students in atleast one club(union) : " << atleast << endl;
cout << "students in neither club (U-union) : " << neither << endl;
}

```

```

enter total no. of students :
200
enter no. of students in sports club :
120
enter no. of students in music club :
90
enter no. of students in both clubs:
50
students : 200
students in sports club : 120
students in music club : 90
students in both clubs : 50
students in only sports club (A-B) : 70
students in only music club (B-A) : 40
students in atleast one club(union) : 160
students in neither club (U-union) : 40

-----
(program exited with code: 0)
Press return to continue

```